

Systems programmer.

Education

2020–2022

Bachelor of Science, The University of Melbourne

Majored in Computing and Software Systems. Graduated with *First-Class Honours*. Relevant subjects include *Algorithms and Data Structures*, *Computer Systems*, *Probability*, and *Statistics*.

Skills

Technical

- Fundamentals** Deep understanding of hardware and software architecture — all the way from CPU architecture to operating systems implementation.
- Programming** Experience in a variety of languages and paradigms, including object-oriented and functional practices. Most fluent with C++ and Rust, with working knowledge of Python and Java, plus others. Very capable of learning on the fly and picking up new technologies quickly.
- Security** Understanding of security best-practices and fundamentals with common services (e.g., Apache and nginx), with experience in implementation of real systems.

Communication

- Written** Strong written communication skills both in an informative and argumentative style.

Experience

References available upon request.

2022–

Software Engineer, IMC Trading

LOW-LATENCY
ALGORITHM DEVELOPMENT

Worked across a variety of systems at various levels of a mature low-latency trading stack:

- ▷ Joined working on low-latency execution and connectivity on a growing desk.
- ▷ Transitioned to baseval algorithm development and research support.
- ▷ Now building a new execution strategy framework, designed from scratch for D1.

2021–2022

Software Engineer Intern, IMC Trading

CONCURRENCY
NETWORKING
PERFORMANCE

Worked on microsecond-sensitive concurrent infrastructure, delivering a project which provides monitoring of a high-performance IPC system with near-zero overhead.

Names are hyperlinked to the relevant repository.

–	Short Circuit , Web Server
SERVICES NETWORKING HTTP	A high-performance web server for Linux using the new <code>io_uring</code> asynchronous I/O interface. ▷ Unix network programming. ▷ HTTP parsing and implementation. ▷ Performance profiling and optimization.
–	liba3 , Utility Library
C LIBRARY DESIGN DATA STRUCTURES	A lightweight utility library in C implementing useful features and data structures absent from the standard library. ▷ More ergonomic strings. ▷ Type-generic hash table, cache, list, and priority queue. ▷ Growable buffer. ▷ Object pool allocator.
–	4cc , C Compiler
COMPILERS PARSING	A C compiler, in C. The ultimate goal is to self-host, but it can already compile a couple “real” programs (see the test suite).
–	Syzygy , Kernel
OPERATING SYSTEMS COMPUTER ARCHITECTURE	A kernel implemented in Rust, currently featuring physical and virtual memory management, an initramfs, and interrupt handling. Presently working on multitasking. ▷ Kernel and bare-metal programming. ▷ <code>no-std</code> Rust programming. ▷ <code>x86_64</code> architecture and assembly.